

To address your concerns, we provide the corresponding theoretical analysis as follows:

Assume that the dataset $\mathcal{D}$ contains a latent true objective reward distribution $r^*(\mathbf{x}, \mathbf{y})$, along with a portion of labels containing noise or flipped labels. Based on the generalization properties of deep neural networks under KL divergence constraints, the model preferentially fits the global reward manifold rather than isolated label noise.

- The positive policy $\pi_{\theta}^{\text{positive}}$ implicitly fits the true reward manifold, so its margin asymptotically converges to the true objective margin Δr^* :

$$\Delta \hat{r}_{\text{pos}} \rightarrow \Delta r^*.$$

- The inverse policy $\pi_{\theta}^{\text{inverse}}$ is trained on globally flipped labels and converges to the inverted reward manifold:

$$\Delta \hat{r}_{\text{inv}} \rightarrow -\Delta r^*.$$

Substituting the above asymptotic limits into the formula for R_{AD} , we obtain:

$$\lim_{\theta \rightarrow \theta^*} R_{\text{AD}} = \frac{1}{\beta} \left(\Delta r^* - (-\Delta r^*) \right) = \frac{2}{\beta} \Delta r^*.$$

This analysis demonstrates that R_{AD} is strictly proportional to the true, objective quality difference Δr^* , and is completely independent of the empirical labels in $\mathcal{D}$ (regardless of whether they contain noise).

Therefore, our filtering function $\phi(R_{\text{AD}}; \tau)$ is built upon solid theoretical bounds:

- True positive preference ($\phi = 1$): When $\Delta r^* > \frac{\beta\tau}{2}$, $R_{\text{AD}} > \tau$. The sample is correctly labeled and should be retained.
- Label noise/polarity flip ($\phi = -1$): When the empirical label is incorrect (i.e., the true $\Delta r^* < -\frac{\beta\tau}{2}$), $R_{\text{AD}} < -\tau$. The transformation function $\mathcal{F}$ spontaneously corrects this label at the mathematical level.
- Ambiguous margin ($\phi = 0$): When $|\Delta r^*| \leq \frac{\beta\tau}{2}$, $|R_{\text{AD}}| \leq \tau$. In other words, these samples may exhibit different biases under different contexts and are therefore not suitable as high-quality preference samples for training.